MARITIME POWER RESEARCH · CYBER DEFENSE · STRATEGIC STUDIES

# Cyber power as a structural variable of maritime power

*A normalized multiplicative model for multidomain assessment of maritime power*

$$MP_{norm} = [(\ MI \times NP\ ) \times SW \times MS^{\gamma} \times CP^{\alpha}\ ]^{1/n_{eff}}$$

$n_{eff} = 3 + \gamma + \alpha$ · $MP_{norm} \in (0, 1]$ · weighted geometric mean (OECD–JRC, 2008; UNDP, 2010)

**Diego E. Cabuya-Padilla · Sergio Uribe-Cáceres · Carlos A. Castaneda-Marroquin**

Naval Academy of Strategic Studies, Colombian Navy · War College · "Almirante Padilla" Naval Cadet School

Manuscript under review – IEEE Access · diegocabuya.github.io/Cyber_Power_and_Maritime_Power/simulator/simulator.html

OUTLINE

# Roadmap of the presentation

| | | |
|---|---|---|
| 1 | **Problem** | A sound formula with a structural gap; question and hypothesis |
| 2 | **Objectives** | General objective and five specific objectives |
| 3 | **Theoretical framework** | Power and its instruments · the maritime power tradition · Booth · the institutional framework · gaps |
| 4 | **Method** | Analytical-conceptual approach in four stages |
| 5 | **Results** | The two-level formula, its verification, its robustness and the Colombian illustration |
| 6 | **Discussion** | Contributions, theoretical and policy implications, limitations and agenda |
| 7 | **Conclusions** | What the article establishes |

THE RESEARCH PROBLEM

# A sound formula with a structural gap

## THE CLASSICAL FORMULATION

$$MP = MI \times NP$$

*Maritime power = maritime interests × naval power*

**Solís Oyarzún (1997)** – Latin American systematization
**Uribe Cáceres et al. (2016)** – adoption for Colombia
**National ocean policy (CCO, 2018)** – state policy: Medium Oceanic Power
**Naval Development Plan 2042 (ARC, 2022)** – currency in naval planning

## THE CONCEPTUAL GAP

### Cyberspace does not appear as a structural domain of maritime power

Navigation, communications, traffic management, ports and surveillance depend on interconnected digital infrastructures whose vulnerability can affect the exercise of maritime power without physical confrontation.

Nor is strategic will formalized as a variable of the system, nor is maritime security disaggregated.

*The article updates the formula – it does not replace it – in continuity with the tradition of the field.*

EMPIRICAL EVIDENCE

# Cyberspace already degrades maritime power

## USD 300 M

**Cyberattack on Maersk (2017)**

Operations disrupted and losses estimated at USD 300 million.

*Wesley et al. (2019); Abbatemarco et al. (2024)*

## +900 %

**Breaches in operational technology**

A single compromised system can give access to the other onboard systems.

*Akpan et al. (2022)*

## AIS · ECDIS

**Critical maritime systems**

Documented vulnerabilities in automatic identification and electronic charts.

*Martínez et al. (2024)*

## 2017 → 2021

**Resolution MSC.428(98) – IMO**

Cyber risk must be addressed in safety management systems.

*IMO (2017)*

**Colombia:** ~988,000 km² of jurisdictional maritime territory and eighteen recognized maritime interests (ARC, 2020, 2022; CCO, 2021).

QUESTION AND HYPOTHESIS

# What the article answers

**RESEARCH QUESTION**

*How should the conceptual expression of maritime power be reformulated to incorporate cyberspace as a strategic domain?*

**HYPOTHESIS**

Cyber power is a structural variable whose omission yields an incomplete representation of state capabilities and vulnerabilities.

OBJECTIVES

# General and specific objectives

## GENERAL OBJECTIVE

To develop an extended formulation of maritime power that integrates cyber power, strategic will and maritime security, with international theoretical scope and an analytical illustration for Colombia.

## SPECIFIC OBJECTIVES

1. Review, through a documented and reproducible search (2014–2026), the antecedents of the formulation and measurement of maritime power and cyber power.
2. Critically analyze the classical formulation MP = MI × NP to establish its content and analytical limits.
3. Build an extended two-level formulation, normalized and justified in the literature, with operational boundaries between components.
4. Verify its logical-mathematical properties, its sensitivity and its robustness to uncertainty.
5. Illustrate its application to the Colombian case and identify the components that concentrate the largest gaps.

FRAMEWORK · POWER AND ITS INSTRUMENTS

# Where maritime power sits

**Power:** the ability to affect others to obtain the outcomes one wants; it is stated only with its scope and domain (Nye, 2010).

**Instruments of national power** (NATO, 2023)

| Diplomatic | Information | **Military** | Economic |
|---|---|---|---|

**Maritime interests** – economic, social and political: what the state seeks at sea.

**Naval power** – the expression of military power at sea: its military character gives meaning to its other functions (Booth, 1977).

**Maritime power** = the capacity to create, develop, exploit and defend maritime interests, in peace and in conflict (CCO, 2018): it combines naval power with maritime interests.

**Cyber power** – obtains outcomes within and beyond cyberspace (Nye, 2010): **it cuts across both components**; that is the premise of the model.

FRAMEWORK · THE TRADITION

# A 130-year tradition, and its next link

**1890**
**Mahan**
Six conditions of sea power

**1911**
**Corbett**
Control of sea communications

**1977**
**Booth**
Trinity: military, diplomatic, policing

**1997**
**Solís Oyarzún**
MP = MI × NP

**2007**
**Till**
System of interconnected elements

**2016**
**Uribe Cáceres et al.**
Adoption for Colombia

**2026**
**This proposal**
Extended, normalized formula

## THREE LITERATURES THE ARTICLE BRINGS TOGETHER

**Maritime power**
Mahan · Corbett · Castex · Booth · Till · Solís Oyarzún · Uribe Cáceres

**Maritime security**
Bueger & Edmunds (2024) – IC · TE · BC

**Cyber power**
Nye (2010) · Bebber (2022) · Cabuya-Padilla et al. (2024, 2025)

FRAMEWORK · BOOTH (1977)

# What navies do, and what cyberspace changes

## THE TRINITY OF NAVAL FUNCTIONS

Unified by the use of the sea:

- **Military** – the base of the trinity
- **Diplomatic** – foreign policy short of the use of force
- **Policing** – sovereignty over one's own maritime frontiers

They do not weigh equally: each state weighs them by its interests → the model weights are calibrated by state.

## LOSS-OF-POWER GRADIENT

**Naval power**: decays with distance until it ceases to be effective (Booth, 1977, p. 120).



**Cyberspace**: instantaneous global reach (NATO, 2025, para. 2.25).




*Cyber power does not decay with distance: hence it enters as a factor of the whole system, not as part of naval force.*

FRAMEWORK · COLOMBIAN INSTITUTIONAL FRAMEWORK

# From ocean policy to the naval plan

**PNOEC 2016-2030 (CCO, 2018)**

State policy: make Colombia a **Medium Oceanic Power**. Frame of the actions of all sectors and input to the National Development Plan and the sectoral plans.

**Position** **Political will** **Maritime interests**

↓

**NAVAL DEVELOPMENT PLAN 2042 (ARC, 2022)**

*"To protect and promote maritime and fluvial interests through naval power"*

**18 maritime interests** (CCO, 2021). The 13 of the PNOEC included naval power; the list of 18 excludes it and avoids double counting in MI × NP.

**Institutional definition**: maritime power as "the sum" of interests and naval power (CCO, 2021). The model adopts the product: a sum would let the absence of a component be offset.

**No cyber mention** in the PNOEC or in the two documents that define the maritime interests (ARC, 2020; CCO, 2021).

FRAMEWORK · STATE OF THE ART

# Four gaps no antecedent resolves

**1 Cyber power is not formalized**

None of the 812 screened records formalizes it as a structural variable of a maritime power measure.

**2 Strategic will stays outside the system**

It is treated as a driver of naval power, not as a condition of the whole system.

**3 Maritime security not disaggregated**

Without the threat dimensions of the contemporary agenda or cyberspace.

**4 Existing measures are additive**

A deficit in one component is fully offset by a surplus in another.

**Documented search** (OpenAlex + Scopus, 13-oct-2026, 2014–2026): 1,045 records → 812 unique → 13 measure maritime or naval power → **0 include cyber power**. IMEMO, Chairil et al., Crisher and Souva, Zhang et al., Vargas-Suárez et al.: all additive.

METHOD

# Analytical-conceptual approach, in four stages

**Conceptual formalization precedes quantitative operationalization** – the same tradition as Mahan, Corbett, Till and Solís Oyarzún.

1

STAGE 1

**Documented search**

Four queries in OpenAlex and Scopus; records and screening published.

2

STAGE 2

**Critical analysis**

Content and limits of the classical formula in its sources.

3

STAGE 3

**Building the formula**

Two levels; justified functional forms; normalization by $n_{eff}$.

4

STAGE 4

**Illustration: Colombia**

MS from Vargas-Suárez et al. (2021); other values illustrative.

**Verification ≠ calibration ≠ external validation** (Sargent, 2011). The model is verified; calibration and external validation are left as an agenda, with a stated protocol.

CENTRAL RESULT · LEVEL 1

# The extended formula of maritime power

$$MP_{norm} = [(\ MI \times NP\ ) \times SW \times MS^{\gamma} \times CP^{\alpha}\ ]^{1/n_{eff}}$$

$$n_{eff} = 3 + \gamma + \alpha \rightarrow MP_{norm} \in (0, 1] \cdot \omega_{MI} = \omega_{NP} = \omega_{SW} = 1/n_{eff},\ \omega_{MS} = \gamma/n_{eff},\ \omega_{CP} = \alpha/n_{eff}$$

**MI**
**Maritime interests**
Purpose: what the state seeks at sea

**NP**
**Naval power**
Instrument: force, position and MDA

**SW**
**Strategic will**
Multiplier of the whole system

**MS**
**Maritime security**
Condition of the environment; weight γ

**CP**
**Cyber power**
Structural variable; weight α

Keeps the classical core. **Limited substitutability**: a deficit is only partly offset, and the collapse of any component cannot be offset (OECD–JRC, 2008; Greco et al., 2019).

RESULTS · LEVEL 2

# Disaggregation of the five components

## MI

$$MI = \Sigma\, w_i \cdot MI_i$$

Linear weighted sub-index: interests are substitutable and weights act as substitution rates. Colombia: n = 18 (CCO, 2021).

## NP

$$NP = F \times P \times MDA^{\mu}$$

Force (with tactical-operational cyber defense), position and maritime domain awareness, raised to the level of F × P.

## SW

$$SW = (PW^{w_1} \cdot IW^{w_2} \cdot SocW^{w_3})^{1/\Sigma w}$$

Political, institutional and social will; geometric mean: none offsets the absence of another.

## MS

$$MS = (IC^{a_1} \cdot TE^{a_2} \cdot BC^{a_3})^{1/\Sigma a}$$

Interstate conflicts, terrorism and blue crime: measures the incidence of threats, not capabilities.

## CP

$$CP = \min\{1, (\beta_1 CS_m + \beta_2 CD_m + \beta_3 RC_m)(1 + \delta \cdot CDip)\}$$

Cybersecurity, strategic cyber defense and recovery capacity, amplified by cyber diplomacy.

*Explicit operational boundaries: each indicator is assigned to a single component, to avoid double counting.*

RESULTS · THE CENTRAL CONTRIBUTION

# Cyber power, inside

**$CS_m$** **Maritime cybersecurity** – Controls, threat analysis and information sharing on the IT/OT assets of the domain.

**$CD_m$** **Strategic maritime cyber defense** – National, non-executional capacity: governance, cooperation, doctrine, science, education and deterrent posture.

**$RC_m$** **Maritime recovery capacity** – Speed of restoration: $RC_m = \min\{1, t_{ref}/t\}$.

**CDip** **Cyber diplomacy** – Amplifier only: never reduces CP; saturation keeps CP ≤ 1.

## α

### WEIGHT OF CYBER DEPENDENCE

**Proposition 1:** $MP_{norm} = G^{1-w} \cdot CP^{w}$, $w = \alpha/n_{eff}$.

Raising CP always raises the index; raising α raises it only if CP > G. Concavity holds for **any** $\alpha > 0$.

Colombia: G = 0.464 > CP = 0.40 → a larger α lowers the index (0.464 → 0.450).

*Cyber maturity is read on CP itself, in five bands: incipient, basic, consolidating, advanced, consolidated.*

RESULTS · INTERPRETATION

# A five-level analytical scale

| | | |
|---|---|---|
| **High** | **0.81 – 1.00** | Consolidated multidomain projection |
| **Significant** | **0.61 – 0.80** | Regional power with developed capabilities |
| **Moderate** | **0.41 – 0.60** | Developing power with identifiable gaps<br>★ **Colombia: $MP_{norm}$ = 0.455** |
| **Low** | **0.21 – 0.40** | Incipient capability with structural vulnerabilities |
| **Critical** | **0.00 – 0.20** | Structural maritime vulnerability |

**FOUR USES**

- Diagnosis of the current level
- Target level for planning
- Comparison among middle powers
- Scenarios in the simulator

*Five equal intervals of the normalized range: conventions of the article, not thresholds estimated from data.*

RESULTS · VERIFICATION

# Three verification stages

**A — Logical-mathematical properties**

Collapse, monotonicity, concavity for any $\alpha, \gamma > 0$, bounds. 300,000 draws; maximum deviation $1.2 \times 10^{-10}$.

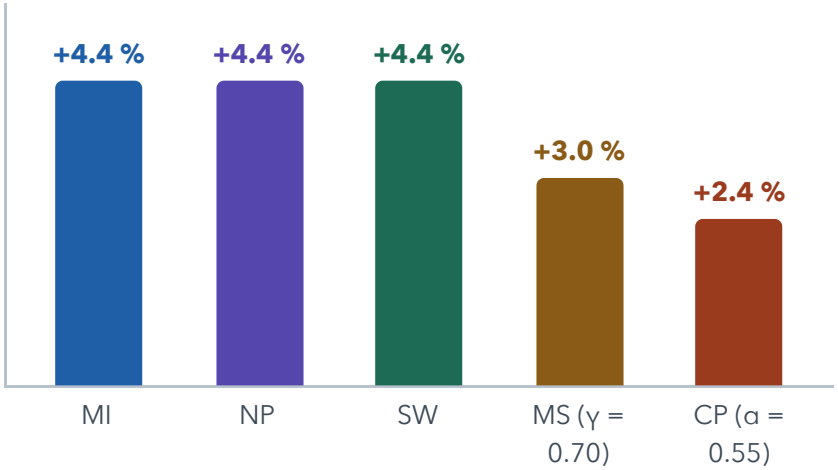
**B — Sensitivity (+20%)**

The elasticity of each component equals its normalized weight $\omega_k$.

**C — Constructed profiles**

Critical ≈ 0.19 · reference ≈ 0.52 · consolidated ≈ 0.85: discriminates along the scale.

## $\Delta MP_{norm}$ for a 20% improvement




*Cyber power has the lowest proportional response ($\omega_{CP} = 0.129$). Sensitivity is not efficiency: comparing investments requires costs.*

RESULTS · UNCERTAINTY AND ROBUSTNESS

# How robust is the result for Colombia?

| | Components ± 0.10 | Components ± 0.20 |
|---|---|---|
| **Median $MP_{norm}$** | 0.464 | 0.452 |
| **P5–P95 band** | [0.406, 0.524] | [0.346, 0.557] |
| **Moderate / Low** | 96.6 % / 3.4 % | 78.1 % / 21.0 % |
| **$S_{Ti}$ of MS · γ** | 0.204 · 0.361 | 0.325 · 0.163 |
| **$S_{Ti}$ of α** | 0.035 | 0.025 |
| **MS is the lowest** | 98.0 % | 76.8 % |

Monte Carlo with Sobol indices (Saisana et al., 2005; Saltelli et al., 2010); α and γ in [0.10, 1]. Ranges are the authors' assumptions, not estimates.

**1 · Moderate is the most likely level**

Robust to moderate uncertainty; with wide ranges, Low appears in 21% of the draws.

**2 · MS and γ dominate the variance**

Measuring maritime security and calibrating γ are the priorities of the index; α is the least influential factor.

**3 · The aggregation rule does not change the level**

Arithmetic mean 0.454 · unweighted geometric 0.433 · $MP_{norm}$ 0.455: all three Moderate.

RESULTS · ANALYTICAL ILLUSTRATION

# Colombia: what the model decomposes

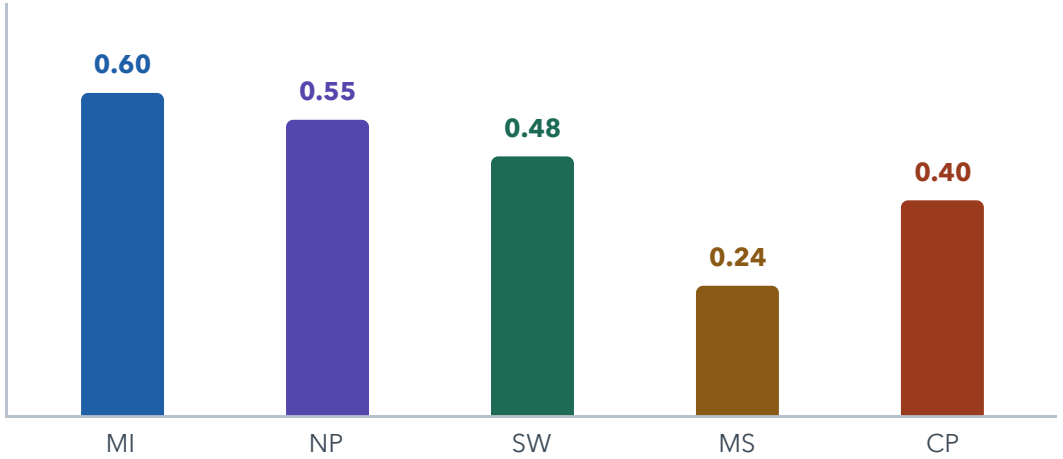
$MP_{norm}$

**0.455**

**MODERATE LEVEL**

Only MS comes from an external source: rank 75 of 99 in maritime security (Vargas-Suárez et al., 2021), converted into a position percentile, (99 − 75)/98 ≈ 0.24. The other values and the parameters are illustrative.

**Reference values by component (α = 0.55; γ = 0.70)**




Maritime security is the lowest component and the one with the largest marginal return; cyber power is the second lowest, at the top of the *basic* band.

*An illustration, not a validation: the value is not compared with the Vargas-Suárez et al. index, which is on another scale.*

RESULTS · STRATEGIC IMPLICATION

# Is cyber power the most efficient investment vector? It depends on costs

### Absolute marginal return $\partial MP_{norm}/\partial X_k$

| Component | Return |
|---|---|
| MS – Maritime security | 0.312 |
| SW – Strategic will | 0.223 |
| NP – Naval power | 0.195 |
| MI – Maritime interests | 0.178 |
| CP – Cyber power | 0.147 |

With the Colombian values, cyber power ranks last in marginal return.

### Corollary 1 – cost condition

$$c_k/c_{CP} > (\omega_k/X_k) \,/\, (\omega_{CP}/CP)$$

Cyber power would be the most efficient vector only if raising each other component by one unit cost at least:

| 1.21× | 1.32× | 1.52× | 2.12× |
|---|---|---|---|
| MI | NP | SW | MS |

what it costs to raise cyber power. The model contains no costs: this is a testable condition, not a result.

*Sensitivity measures responsiveness, not efficiency: prioritizing an investment requires knowing the relative costs.*

DISCUSSION · CONTRIBUTIONS

# Six contributions

| | | |
|---|---|---|
| 1 | **Cyber power as a structural variable** | With explicit operational boundaries and α as a weight, not a capability. |
| 2 | **Strategic will as a multiplier of the system** | It conditions all of maritime power, not only naval power. |
| 3 | **Disaggregated maritime security** | IC · TE · BC as incidence of threats, with weight γ. |
| 4 | **Normalization, scale and verification** | $MP_{norm} \in (0, 1]$, five levels, Proposition 1 and Corollary 1; verification distinguished from calibration and validation. |
| 5 | **MDA ≠ maritime consciousness** | Technical-operational capability in NP; cultural construct in SW. |
| 6 | **Maritime power within a general theory of power** | Naval power as the expression of military power; maritime interests as the economic, social and political component; cyber power cross-cutting. |

DISCUSSION · IMPLICATIONS

# Two implications beyond the formula

THEORETICAL

## Cyber power escapes the geographic logic of navies

Classical theory distinguishes navies by a reach that decays with distance (Booth, 1977); cyberspace offers instantaneous global reach (NATO, 2025).

**Consequence:** for states with coastal or contiguous-sea navies, cyberspace removes the protection that distance afforded.

POLICY

## Ocean policy already counts will, but not cyber

The PNOEC frames the development plans and treats political will as a variable of its goal (CCO, 2018); neither it nor the documents that define the maritime interests mention the cyber dimension, and the institutional definition is a sum.

**Consequence:** the model offers the structure to incorporate it in the next update and to track progress with an index in which no collapse is offset.

DISCUSSION · LIMITS AND AGENDA

# Stated limitations, an agenda with a protocol

## LIMITATIONS

- Verified, but neither calibrated nor externally validated.
- In the illustration, only MS comes from an external source.
- Static model; the enabling effect of cyber power on NP and MS is argued but not represented.
- Robust to moderate uncertainty, not to large uncertainty.

## CALIBRATION AND VALIDATION PROTOCOL (NATIONAL INDEX)

1. **Expert calibration:** AHP panel on the five components; γ and α as weight ratios.
2. **Independent inputs:** published sources (ITU, 2024; NCSI; Wild and Fraser, 2026).
3. **Uncertainty:** Monte Carlo with each expert's weights and each source's error.
4. **Comparison:** rank correlation with Vargas-Suárez et al. (2021), with the criterion fixed beforehand.

Also: regional extension and dynamic modeling with SERDUX-MARCIM.

CONCLUSIONS

# Maritime power is also exercised in cyberspace

**01** The classical formula remains valid but is no longer sufficient: the proposal updates it while keeping its multiplicative core.

**02** Cyber power, strategic will and maritime security enter as structural variables, with limited substitutability: the collapse of any component cannot be offset.

**03** Maritime power combines naval power – the expression of military power at sea – with maritime interests; cyber power cuts across both and does not decay with distance.

**04** Colombia is illustrated at the Moderate level ($MP_{norm} \approx 0.455$), with maritime security as the largest gap; this is not a definitive measurement.

**05** Colombian ocean policy already treats will as a variable but does not mention cyber: the model offers the structure to incorporate it.

**06** Whether cyber power is the best investment depends on relative costs (Corollary 1); the national index is the next stage.

**Diego E. Cabuya-Padilla · Sergio Uribe-Cáceres · Carlos A. Castaneda-Marroquin**

Open-access interactive simulator: diegocabuya.github.io/Cyber_Power_and_Maritime_Power/simulator/simulator.html

*Thank you*